

The Hadronic Topological Architecture: Simulating Composite Quantum Identity

Version 2.0 — Revised & Scientifically Deepened | June 2026

Classification: Tardigradia Game Engine | Physics Deep Dive Series | TGPU v2.0 **Particle Class:** Hadrons (composite) | Baryons: qqq | Mesons: $q\bar{q}$ | Exotica: $qq\bar{q}\bar{q}$, $qqq\bar{q}q$ | Strong-force bound
Source: Extends v1.0Hadrons Deep Dive Request.pdf (V1.0)

Δ Change Log V1.0 → V2.0

Parameter	V1.0	V2.0	Source
LHCb exotic hadrons	No exotica in V1.0	V2.0: LHCb 2022-2024: Tcc+ (doubly-charmed tetraquark), 3 pentaquarks, $\chi_{c1}(3872)$ exotic confirmed. V2.0 adds EXOTIC_HADRON category with quark content array up to 5 quarks. Charm quark content tracked separately for tetraquark formation	LHCb exotic hadrons 2022-2024
QCD lattice masses	Experimental masses only	V2.0: BMW Lattice QCD (2021-2024): proton 938.3 MeV (matches exp. 0.5%), neutron 939.6 MeV, Omega baryon 1672.5 MeV. All masses now derived from first-principles α_s and quark masses. In simulation: mass lookup from lattice table rather than experimental PDG	BMW Lattice QCD 2024

Parameter	V1.0	V2.0	Source
Nucleon sigma term update	Not covered	V2.0: $\sigma_{\pi N}$ (pion-nucleon sigma term) = 38 ± 4 MeV (FLAG 2024 lattice average). Critical for dark matter nuclear scattering cross-section. In simulation: $\sigma_{\pi N}$ determines DARK_MATTER_INTERACTION rate with nucleons	FLAG 2024; LQCD review
Hadron string breaking simulation	String tension noted	V2.0: Lattice QCD string breaking: $V(r) = \sigma * r$ until $r = r_{\text{break}} \sim 1.2$ fm, then quark-antiquark pair creation. In simulation: LUND string fragmentation model — string breaks with probability $P = \exp(-\pi m_q^2 / \sigma)$ per unit length per unit time	Lund string model; Altarelli review

1. The One-Particle Universe Framework — V2.0 Update

The foundational architecture established in V1.0 — Worldline Monism, the “Origami Universe” metaphor, and the Object-Oriented ontology distinguishing intrinsic from extrinsic properties — is **fully preserved** and remains physically rigorous. V2.0 layers NEW discoveries from 2022-2026 atop this framework.

1.1 WebGPU Compute Architecture (V2.0 Core Infrastructure)

V1.0 designed for WebGL/Three.js on the CPU-side particle update loop. V2.0 upgrades to WebGPU (available in Chrome 113+ May 2023, Firefox 2024):

```
// WGSL Compute Shader - V2.0 Particle Integration
@binding(0) @group(0) var<storage, read_write> particles: array<ParticleState>;

struct ParticleState {
    position: vec4f,
    momentum: vec4f,
    spin: vec2f,           // Spinor components
    berry_phase: f32,      // Geometric phase accumulated
    proper_time: f32,      // tau - worldline parameter
    color_charge: u32,      // Packed color in 2 bits (R=0,G=1,B=2)
    quantum_numbers: u32,  // Packed: isospin, strangeness, charm, beauty
    decay_lifetime: f32,
```

```

    entanglement_id: i32, // -1 = unentangled; >=0 = entanglement group
}

@compute @workgroup_size(64)
fn updateParticles(@builtin(global_invocation_id) id: vec3u) {
    let i = id.x;
    if (i >= arrayLength(&particles)) { return; }

    var p = particles[i];

    // Feynman-Schwinger proper time evolution
    let dtau = uniforms.dt / p.mass; // proper time step

    // Update worldline parameter
    p.proper_time += dtau;

    // Berry phase accumulation (topological phase)
    p.berry_phase += dot(uniforms.gauge_field, p.momentum.xyz) * dtau;

    // Lorentz force (for charged particles)
    let F = cross(uniforms.B_field, p.momentum.xyz) * p.charge_sign;
    p.momentum.xyz += F * uniforms.dt;
    p.position.xyz += p.momentum.xyz * uniforms.dt / p.energy;

    particles[i] = p;
}

```

1.2 Updated Physical Constants (CODATA 2022)

Constant	V1.0 Value	V2.0 Value (CODATA 2022)
m _e	9.10938356×10 ⁻³¹ kg	9.1093837139×10 ⁻³¹ kg
e	1.60217663×10 ⁻¹⁹ C	1.602176634×10 ⁻¹⁹ C (exact)
ħ	1.054571817×10 ⁻³⁴ J·s	1.054571817×10 ⁻³⁴ J·s (exact)
c	2.99792458×10 ⁸ m/s	2.99792458×10 ⁸ m/s (exact)
α	1/137.036	1/137.035999084 ±0.000000021

2. V2.0 Enhanced Data Structures

Extending V1.0 struct definitions with 2022-2026 discoveries:

```

// V2.0 Universal Particle State – extends V1.0
struct ParticleInstanceV2 {
    // Inherited from V1.0
    double proper_time; // tau worldline parameter (meter)
    glm::dvec3 position; // 3-space vector
    glm::dvec3 momentum; // 3-momentum (relativistic)
    float spin_z; // +0.5 or -0.5 (or eigenvalue)
    float berry_phase; // Accumulated Berry/Aharonov-Bohm phase

    // V2.0 ADDITIONS
}

```

```

float entanglement_r;          // Reduced density matrix purity 0-1
int entanglement_partner_id;   // -1 = none; global particle ID
float flavor_oscillation_phase; // For neutrinos/kaons
uint32_t color_bits;           // 3-bit: R(b0),G(b1),B(b2)
float decay_probability_dt;     // P(decay in dt) for unstable particles
float vacuum_alignment;        // Order parameter for condensate mode

// WebGPU Buffer packing
// Total: ~100 bytes/particle * 10^6 particles = 100 MB GPU memory
float _pad[2];                 // Align to 16 bytes for WGS
};

```

8. V2.0 Benevolence Metric Update

The “Benevolence” concept central to all OEU documents is given a precise V2.0 mathematical definition grounded in 2022-2026 quantum information theory:

Benevolence = Quantum Coherence + Information Preservation + Symmetry Maintenance

$$B(t) = C(t) \cdot F(t) \cdot S(t)$$

Where: - $C(t)$ = l_1 -norm coherence of particle density matrix (Baumgratz et al. 2014) - $F(t)$ = quantum Fisher information / classical Fisher information (ratio ≥ 1) - $S(t)$ = fraction of symmetries unbroken at time t

V2.0 game mechanic: Benevolence score increases when: - Particles maintain long entanglement coherence (high C) - Topological protections are active (Berry phase winding conserved) - Symmetry groups are preserved in interactions (no spurious symmetry breaking)

End of v1.0 Hadrons Deep Dive Request V2.0 | Tardigradia Game Engine SubParticles Series This document supersedes V1.0 in all physics specifications.